

True/False

1. **TRUE** False For $f(x)$ continuous on $[a, b]$, the function $F(x) = \int_a^x f(u)du$ is the unique anti-derivative of f on $[a, b]$ satisfying $F(a) = 0$.
2. **TRUE** False If $f(x)$ is continuous on $[a, b]$, then $F(x) = \int_a^x f(u)du$ is continuous on (a, b) .
3. **TRUE** False We can use the addition differentiation law to prove the addition integration law for indefinite integrals $(\int (f + g) = (\int f) + (\int g))$.

Integration by Parts

Examples

4. True **FALSE** Integration by parts will automatically give the antiderivative of a function.
5. Find $\int \arctan(x)dx$.

Solution: For choosing a u , we should look for logarithms and inverse trig functions first. Then, we should look for polynomials, then trig functions and finally exponential functions. So, we set $u = \arctan(x)$ and set dv as the rest or $dv = 1dx$. Thus, we have that

$$\int \arctan(x)dx = uv - \int vdu = x \arctan(x) - \int \frac{x}{1+x^2}dx.$$

We can solve the last one with a u substitution $u = 1+x^2$ so $du = 2xdx$ and $xdx = \frac{du}{2}$ so

$$\int \arctan(x)dx = x \arctan x - \frac{1}{2} \int \frac{du}{u} = x \arctan x - \frac{\ln(1+x^2)}{2} + C.$$

6. Find $\int \sin(x) \cos(x)dx$.

Solution: We can use u substitution but also use integration by parts. Let $u = \sin(x)$ and $dv = \cos(x)dx$ so $v = \sin(x)$. Thus

$$\int \sin(x) \cos(x) dx = \sin^2(x) - \int \cos(x) \sin(x) dx.$$

So we have that

$$2 \int \sin(x) \cos(x) dx = \sin^2(x) + C \implies \int \sin(x) \cos(x) dx = \frac{\sin^2(x)}{2} + C.$$

7. Integrate $\int 2x^3 \cos(x^2) dx$.

Solution: Let $u = \cos(x^2)$ and $dv = 2x^3 dx$ so $du = -\sin(x^2) \cdot 2x dx$ and $v = \frac{x^4}{2}$. Then

$$\int 2x^3 \cos(x^2) dx = \frac{x^4 \cos(x^2)}{2} - \int -\sin(x^2) x^5 dx.$$

This hasn't been simplified at all so we should try another approach. Instead, first u sub to get $u = x^2$ and $dx = 2x$ so $2x^3 dx = 2x dx(x^2) = u du$. Therefore, we get

$$\int 2x^3 \cos(x^2) dx = \int u \cos(u) du = u \sin(u) + \cos(u) + C = x^2 \sin(x^2) + \cos(x^2) + C.$$

Problems

8. Integrate $\int x \ln x dx$.

Solution: Let $u = \ln x$ and $dv = x dx$ so

$$\int x \ln x dx = \frac{x^2 \ln x}{2} - \int \frac{x^2}{2} \frac{1}{x} dx = \frac{x^2 \ln x}{2} - \int \frac{x}{2} dx = \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C.$$

9. Integrate $\int \frac{\ln x}{x^5} dx$.

Solution: Let $u = \ln x$ and $dv = x^{-5}dx$ so $v = \frac{-x^{-4}}{4}$ and so

$$\int \frac{\ln x}{x^5} dx = \frac{-\ln x}{4x^4} - \int \frac{-1}{4x^4} \cdot \frac{1}{x} dx = \frac{-\ln x}{4x^4} - \int \frac{-1}{4x^5} dx = \frac{-\ln x}{4x^4} - \frac{1}{16x^4} + C.$$

10. Integrate $\int 2x \arctan(x) dx$.

Solution: Let $u = \arctan(x)$ and $dv = 2x dx$ so $v = x^2$. Then

$$\begin{aligned} \int 2x \arctan(x) dx &= x^2 \arctan(x) - \int \frac{x^2 dx}{1+x^2} = x^2 \arctan(x) - \int \frac{x^2+1}{x^2+1} - \frac{1}{x^2+1} dx \\ &= x^2 \arctan(x) - x + \arctan(x) + C. \end{aligned}$$

11. Integrate $\int (\ln x)^2 dx$.

Solution: Let $u = (\ln x)^2 dx$ and $dv = 1 dx$ so $v = x$. Then $du = \frac{2 \ln x}{x} dx$ and

$$\begin{aligned} \int (\ln x)^2 dx &= x(\ln x)^2 - \int \frac{2x \ln x}{x} dx = x(\ln x)^2 - 2 \int \ln x dx \\ &= x(\ln x)^2 - 2 \left[x \ln x - \int x/x dx \right] = x(\ln x)^2 - 2x \ln x + 2x + C. \end{aligned}$$

12. Integrate $\int x(\sin x + \cos x) dx$.

Solution: Let $u = x$ and $dv = (\sin(x) + \cos(x)) dx$ so $du = dx$ and $v = -\cos(x) + \sin(x)$. So

$$\begin{aligned} \int x(\sin(x) + \cos(x)) dx &= x(\sin(x) - \cos(x)) - \int \sin(x) - \cos(x) dx \\ &= x(\sin(x) - \cos(x)) - (-\cos(x) - \sin(x)) + C. \end{aligned}$$

13. Integrate $\int \frac{\ln \sqrt{x}}{\sqrt{x}} dx$.

Solution: Let $u = \ln(\sqrt{x})$ and $dv = \frac{dx}{\sqrt{x}} = x^{-1/2}dx$ so $v = 2x^{1/2} = 2\sqrt{x}$. Then by chain rule, we have $du = \frac{1}{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}dx = \frac{1}{2x}dx$. Thus, we have

$$\int \frac{\ln \sqrt{x}}{\sqrt{x}} dx = 2\sqrt{x} \ln \sqrt{x} - \int 2\sqrt{x} \cdot \frac{dx}{2x} = 2\sqrt{x} \ln \sqrt{x} - \int \frac{dx}{\sqrt{x}} = 2\sqrt{x} \ln \sqrt{x} - 2\sqrt{x} + C.$$

Numerical Integration

14. True **FALSE** For calculating the error bound when using left endpoint method when approximating the integral of f on the interval $[a, b]$, we use $K_1 = f'(a)$.
15. **TRUE** False The error for an integral approximation can be negative.
16. True **FALSE** The error bound gives us what the exact error of using the different approximation techniques are.
17. True **FALSE** The error bounds aren't helpful because they don't give us the exact error.